

Section 2.4

I. Definition of the exponential function b^x

- a. $b \neq 0$
- b. $b^n = \underbrace{b \cdot b \cdot b \cdots b}_{n \text{ times}}, n \text{ natural number, } b^0 = 1$
- c. $b^{-n} = \frac{1}{b^n}, n \text{ natural number}$
- d. $b^{1/n} = \sqrt[n]{b}, n \text{ natural number}$
- e. $b^{m/n} = \left(b^{1/n}\right)^m, m/n \text{ rational number}$
- f. $b^x = \lim_{\substack{m/n \rightarrow x \\ m/n \text{ rational}}} b^{m/n}, x \text{ real number}$

II. Theorem 2.8 Properties of Exponential Functions

III. Graphical Properties ($b > 1, b < 1$)

- a. Domain, Range
- b. Graph
- c. Asymptote
- d. Monotonic

IV. Logarithm: Inverse of Exponential

V. Graphical Properties ($b > 1, b < 1$)

- a. Domain, Range
- b. Graph
- c. Asymptote
- d. Monotonic

VI. Theorem 2.9 Properties of Logarithm Functions

VII. Solving Equations

Examples

VIII. Natural Exponential and Logarithm Function

a.
$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

b.
$$e^x = \exp(x)$$

c.
$$\log_e x = \ln x$$

d. Change of base
$$\log_b x = \frac{\ln x}{\ln b}$$

e. Basic Properties

IX. Modelling

a. Exponential Growth

b. Continuous Compound Interest

Examples